

SANGEETH BIJUKUMAR

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Profile Summary

Full-stack and AI Engineer with hands-on experience building and deploying end-to-end LLM-powered applications, including RAG pipelines and agent-based systems using LangChain and modern AI models. Experienced in Dockerized deployments on AWS (ECS, Fargate) and working with vector databases for scalable AI solutions. Delivered real-world projects in both industry and research settings, including secure DNN watermarking and production-grade AI systems. Focused on building practical, high-impact AI products.

Technical Skills

Programming languages: Python, JavaScript, Java, SQL

Web-Development: React , Express, Node.js , FastApi ,Celery, Redis

Generative AI & AI/ML Stack: NLP, Hugging Face, OpenAi, Claude, LangChain, LangGraph, RAGAS, GaurdRails

Cloud & DevOps: AWS (EC2, S3, ECS, Fargate), Docker, Kubernetes

Databases: PostgreSQL, PGvector, MongoDB, Neo4j

Tools and Technologies: Colab, Anaconda, Git, VS code

Experience/Projects

Stirrup communications consultant

August 2025 – January 2026

Intern

Ernakulam, India

- Built **RAG (Retrieval-Augmented Generation)** and **Agentic RAG** solutions using **Anthropic's Claude models**.
- Experimented with and tested **LightRAG** implementations to compare efficiency, scalability, and response quality across RAG architectures.
- Designed **LLM-driven agents** capable of multi-step reasoning, decision-making, and data interaction through LangChain frameworks.
- Collaborated with the AI team to design and evaluate prompt engineering strategies and improve model performance across multiple tasks

Cyber Security Research lab, CUSAT

December 2024 – May 2025

Intern

Ernakulam, India

- This project was done under the guidance of **Dr Vinod P** and collaboration with **researchers from Italy**.
- Developed a **Python-based watermarking system for deep neural networks** to protect model ownership.
- Watermarked the model by **integrating chaotic sequence(generated using logistic maps)** to the **weights** of the model.
- Improved model security by designing a **detection method** using **machine learning classifiers**.
- Optimized watermarking approach to maintain **98% accuracy** while enhancing robustness against attacks.
- Created a verification method using **genetic algorithm** to verify the model ownership.

AskDocs — AI-Powered Document Intelligence System (Personal Project)

- Built a production-ready multi-agent RAG system with FastAPI, Celery, Redis, and pgvector
- Implemented hybrid retrieval (vector + keyword + RRF fusion) with real-time SSE streaming
- Deployed on AWS S3 with Docker; includes guardrails for prompt injection, PII, and toxicity filtering
- GitHub: <https://github.com/sangeeth40/AskDocs-server>

Workshops/Certifications

- IBM AI Engineering, Coursera
- IOT Essentials for Industry 4.0 by AICTE, CUSAT (January 2024 - February 2024)
- Google Cybersecurity, Coursera
- Python Basics, University of Michigan, Coursera
- Python for Data Science, IBM, Coursera

Education

SCMS School of Engineering and Technology, KTU university

2023 – 2025

Master of Computer Applications

Ernakulam, Kerala

Amritapuri Campus, Amrita university

2020 – 2023

Bachelor of Computer Applications

Kollam, Kerala